

- Data structures are designed to characterize the objects.
- Functions operating on the data of an object are tied together in the data structures.
- Data used is generally hidden and cannot be accessed by external functions.
- Functions help objects to communicate with each other.
- New functions and data can easily be added as per need.
- A bottom up approach is followed during program design.

1.1 Basic Concepts

Before delving into OOP, it is important to be familiar with its concepts. These include:

- Classes
- Objects
- Dynamic Binding
- Message passing

Classes

Classes are used to implement the concept of Abstract Data Types (ADT). A class is a combination of both properties and methods used to manipulate properties. In fact, a class is a blueprint describing the nature of the data structure. For example, consider the case of the class 'student'. There are some common properties shared by all students, such as name, roll, class, address, and marks. Similarly, there might be some methods used to manipulate these properties. However, the values of these properties can differ depending in the student. If we want to use the class 'student', then we need to create instances of this class, also known as objects. Classes are user-defined data types and behave like a built-in programming language.

Objects

Objects are the basic runtime entities in an object oriented system. In fact, it is the instance of a class. An object can be a person, place, bank account or even a table of data that the pro-